

Baoren Liu

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

GPA: 3.92; Bachelor of Science in Mathematics & Computer Science

Expected: May 2028

Activities: SWE @ National Organization for Business and Engineering (NOBE) — Technology Division

Relevant Coursework: Data Structures & Algorithms, Database Systems, Linear Algebra, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, HTML, CSS, JavaScript, TypeScript, Kotlin, Bash

Technologies: React, Next.js, Flask, PyTorch, LangChain, NumPy, Pandas, scikit-learn, ChromaDB, Hugging Face

Tools: Git, CI/CD, PostgreSQL, Vercel, Docker, Google Cloud, Microsoft Azure, Linux, Ollama, Isaac Sim, Codex

EXPERIENCE

HCESC-XR Lab, UIUC

July 2026 – Present

Undergraduate Researcher

Champaign, IL

- Developing an **Isaac Sim** benchmark to evaluate **AI agents'** ability to actively diagnose hidden physical failure causes and recover from manipulation failures
- Building **Python** simulation environments with sensor feedback, inverse-kinematics controller, and observation-action API to support **LLM, VLA, and RL agents**

Mass General Brigham

June 2024 – Present

AI Research Intern

Boston, MA

- Built an end-to-end retrieval-augmented generation pipeline to process, embed, retrieve, and classify evidence across **115M+ proprietary clinical records** in a HIPAA-compliant environment
- Co-authored** a research paper on the system, which achieved a **0.927 F1-score** for dementia identification and is currently under peer review: <https://doi.org/10.64898/2026.01.24.26344477>
- Built a **multi-agent** hospital support prototype using **LangGraph** to orchestrate specialized AI agents
- Automated deployment of the team website using **GitLab CI/CD**, SSH-managed Linux runners, and webhooks, eliminating manual deployment steps

PROJECTS

OtherWise | *SentenceTransformer, FastAPI, PCA, UMAP, Python, Codex*

October 2026

- Built a **discovery system** designed to counter algorithmic filter bubbles by guiding users from familiar interests into new knowledge domains: <https://roboticreaper.github.io/OtherWise/>
- Developed a **semantic exploration engine** using **keyword embeddings** to balance relevance and novelty, surfaced through a Chrome extension
- Created interactive **topic-galaxy** and trail visualizations with **PCA/UMAP** to map discoveries and show how users' interests expand over time

RideList | *Next.js, PostgreSQL, Cloudflare API, Vercel, Google Cloud, Azure, Git* December 2025 – October 2026

- Built and launched a bilingual campus ridesharing app with Next.js, supporting **60+ trips, 75+ messages**, and **7 completed bookings**: <https://www.ridelist.app/>
- Secured communication between Vercel-hosted APIs and a self-hosted database using a custom **certificate authority and mTLS**
- Built privacy-preserving geospatial ride matching with **PostGIS**, and implemented real-time messaging, trip synchronization, and automated PostgreSQL backups using Firebase and Bash scripts on Linux

LHS Schedule | *React, Material UI, Firebase, Netlify*

July 2020 – December 2024

- Built a scheduling app with real-time schedule updates, cross-device synchronization, offline access, PDF schedule importing, and social features
- Supported **1,400+ registered users and 900 daily active users** by parallelizing API requests, caching data with a **service worker**, and deploying on Netlify: <https://lhsschedule.netlify.app/>

AWARDS

1st in Beginner Division; 6th of 131 teams overall, SIGPwny (ACM @ UIUC) Fall CTF Competition – 2025